Special Notice (SN) DARPA-SN-26-55: Virtual-Integrated Twin for Autonomous Lifesaving (VITAL) Proposers Day

DATE: **March 31, 2026**
LOCATION: Executive Conference Center (ECC)
4075 Wilson Boulevard, 3rd Floor
Arlington, VA 22203
REGISTRATION WEBSITE: https://events.sa-meetings.com/VITALProposersDay
REGISTRATION DEADLINE: **March 23, 2026 at 5:00 PM EDT** or when in-person capacity (125) and online capacity (350) is reached, whichever comes first. ***Note: due to these capacity limitations, each participating organization may not register more than 2 participants for in-person, and organizations are encouraged to prioritize participants with technical backgrounds for in-person registration.***
TECHNICAL POC: Roozbeh Jafari, PhD, DARPA/BTO
Email: VITAL@DARPA.mil

The Biological Technologies Office (BTO) of the Defense Advanced Research Projects Agency (DARPA) is hosting a Proposers Day for the potential proposer community in support of the Virtual-Integrated Twin for Autonomous Lifesaving (VITAL) Program. The Proposers Day will be held on **March 31, 2026**, in-person and by webcast for remote participation. Advance registration is required. Presentations and discussions will be broadcast via the webcast. Registrants will have the opportunity to provide 1-2 page teaming profiles, give a short "lightning" talk (as described below), participate in a poster session, and/or participate in private one-on-one sessions with the DARPA government team. There is a limited number of timeslots for private one-on-one sessions and oral presentations, so they must be scheduled upon registration on a first come first served basis. Note, all times listed in this announcement and on the registration website are Eastern Time.

The goals of the Proposers Day include:

1. Introduce the scientific community (industry, academia, and government) to the VITAL program vision, goals and objectives.
2. Explain the general mechanics of a DARPA program and the specific objectives and milestones of this program.
3. Increase efficiency in proposal preparation and evaluation.
4. Encourage and promote teaming arrangements among organizations with the expertise, research facilities, and capabilities needed to execute research and development responsive to VITAL program goals and objectives.

DARPA anticipates releasing the VITAL Program Solicitation (PS) in March, 2026. When released, the PS will be available on https://sam.gov/.

It is expected that VITAL will require strong teaming efforts to successfully innovate and integrate critical technologies necessary to meet the metrics (and associated timelines) of the VITAL program. The Proposers Day will include overview presentations by government personnel, "lightning" talks for potential proposers to highlight technical capabilities or interests in teaming, and an information session to respond to questions posed by participants. Lightning talks must be scheduled via the registration website. Lightning talk presentations will be made accessible to the proposer community. Participants may also submit a teaming profile that includes contact information, technical competencies, relevant facilities, along with the technical competencies and/or facilities needed from potential team performers. As with the lightning talk presentations, these teaming profiles will be made accessible to the proposer community.

DARPA will collect participant technical and contractual questions throughout the event, along with program feedback. Answers to submitted questions will be posted online in the publicly available FAQ document, along with presented materials from the Proposers Day, on VITAL program page at DARPA's website: https://www.darpa.mil/research/programs/virtual-integrated-twin-for-autonomous-lifesaving

During registration, participants will also be able to sign up for short one-on-one sessions with DARPA personnel These short 10-minute sessions will be held during the Proposers Day with in-person participants only. One-on-one sessions must be scheduled through the registration website and will be on a first-come, first-served basis until time constraints are reached.

To maximize the pool of innovative proposal concepts, DARPA strongly encourages participation by non-traditional performers, including small and large businesses, as well as academic and research institutions, including first-time government contractors. Attendance at this event is not a requirement for submission of a proposal or selection for funding. Information relayed during the Proposers Day will be made available on the VITAL program page at DARPA's website:
https://www.darpa.mil/research/programs/virtual-integrated-twin-for-autonomous-lifesaving

Program Objective and Description:
The VITAL program aims to develop continuously updating computational models of the cardiovascular system that integrate patient data with biological physics to predict outcomes in real time. The vision of VITAL is to enable future providers and clinicians to assess treatment options for acute and chronic pathologies.

VITAL plans to establish a foundation for causal, prediction-driven decision support using high-fidelity (HF) digital twins that explicitly represent the physical, biochemical, and anatomical dynamics governing cardiovascular physiology. HF models will be developed and benchmarked as the baseline predictive reference across static, chronic, and acute regimes. For each regime, the program intends to quantify computational speed, forecast horizon, predictive accuracy, uncertainty bounds, data requirements, sensitivity to parameter estimation, segmentation error, and measurement sparsity. A core program outcome is a rigorous characterization of HF model capability limits, providing evidence-based guidance on when HF models are suitable as foundational technology and where fundamental limitations remain.

While HF models provide the strongest mechanistic fidelity and causal interpretability, reduced-order models (ROMs) derived from HF dynamics are better suited for real-time responsiveness and continuous updating, particularly when coupled to streaming sensors. Through systematic verification, validation, and uncertainty quantification, VITAL aims to determine when and where different modeling approaches—HF models, ROMs, and measurement-driven inference—are reliable and appropriate.

VITAL plans to establish an Image-to-Physics-to-Twin pipeline that automatically integrates multimodal clinical imaging—such as magnetic resonance imaging, computed tomography, computed tomography angiography, and ultrasound—with sparse physiological and biochemical measurements to construct patient- or archetype-specific high-fidelity digital twins. Segmented anatomy is translated into vascular networks, organs, and injury geometries spanning chronic and acute conditions. Localized three-dimensional injury-site solvers capture bleeding and impedance dynamics, while whole-body 0-dimensional and 1-dimensional physiology models propagate global responses such as shock.

Finally, VITAL will pursue a systematic reduction of HF models into fast, continuously updateable, mechanism-preserving surrogates using artificial intelligence techniques that are explicitly physics- and physiology-aware. These ROMs are trained against HF dynamics and structured to retain vascular coupling and injury–treatment feedback, enabling continuous state and parameter estimation under sparse and noisy sensing.

VITAL is a two-phase program. Phase 1 establishes technical credibility through the development of HF models and rigorous quantification of their performance limits. Phase 2 evaluates the performance tradeoffs associated with HF-to-ROM transitions, including scalability, real-time execution, and intervention-forecasting accuracy. DARPA strongly encourages teaming to ensure the breadth of expertise required to achieve VITAL's objectives.

Teaming Profiles:
DARPA strongly encourages teaming to ensure the expertise and capabilities necessary to meet program goals. Interested parties are invited to augment their Proposers Day participation by submitting a 1-2 page team profile (template is provided on the registration website) that, at a minimum, includes the following:

- Contact information, including name, organization, email, telephone number, mailing address, and website.
- Brief description of the proposer's technical competencies and relevant facilities.
- Desired technical competencies and facilities needed from other potential team performers, if applicable.

All profiles must be emailed to VITAL@darpa.mil no later than **5:00 PM EDT on March 23, 2026**. Following the deadline, a consolidated list of profiles will be sent via email to all registrants. Furthermore, for proposers who were unable to attend the event, the teaming profiles will be made accessible via email request. Profiles that do not use the provided teaming profile template or exceed the two-page limit will not be accepted. It is the responsibility of the submitter to assure all information contained in the teaming profile is non-proprietary, unclassified, and otherwise acceptable for public release. Profile submitters consent to distribution amongst event registrants. Specific content, communications, networking, and team formation are the sole responsibility of the participants. Neither DARPA nor DoD endorses the information and organizations contained in the consolidated teaming profile document, nor does DARPA or DoD exercise any responsibility for improper dissemination of the teaming profiles.

Oral Presentations (Lightning Talks):
Attendees may be afforded the opportunity to give a brief (3 minute) oral presentation outlining their interests and capabilities. Interested attendees are invited to submit one Microsoft Powerpoint slide with no videos or animations that summarizes their interests and capabilities. A lightning talk template will be provided on the registration website.

The purpose of these presentations is to facilitate teaming discussions among the attendees. Oral presentations will be restricted to one slide per presenter and must be submitted to VITAL@darpa.mil no later than **5:00 PM EDT on March 23, 2026**. Upon registering, attendees must indicate whether they would like to give an oral presentation. Due to limited availability, DARPA does not guarantee that these requests will be fulfilled. Lightning talk submissions will be accepted on a first-come, first-served basis until time constraints are reached. It is the responsibility of the submitter to ensure that all information contained in the lightning talk is non-proprietary, unclassified, and otherwise acceptable for public release. Neither DARPA nor DoD exercise any responsibility for improper dissemination of the lightning talks.

Poster Session:
In-person attendees will be afforded the opportunity to prepare posters describing areas of interest and capabilities. The purpose of these presentations is primarily to facilitate teaming discussions among the attendees; however the VITAL Program Manger will also participate in the poster session. Upon registering, attendees may indicate whether they would like to present a poster. Due to limited availability, DARPA does not guarantee that these requests will be granted. Digital versions of posters must be submitted to VITAL@darpa.mil for approval no later than **5:00 PM EDT on March 23, 2026**. DARPA will notify attendees if their poster has been approved for presentation no later than 5:00 PM EDT on March 25, 2026. Attendees are responsible for bringing the preapproved posters to the Proposers Day.

Registration Information:
PLEASE NOTE: Registration closes on **March 23, 2026, at 5:00 PM EDT.**
Participants must register in advance through the registration website. Registration is free but limited by in-person capacity (125) and online capacity (350), so early registration is recommended. ***Note: due to these capacity limitations, each participating organization may not register more than 2 participants for in-person, and organizations are encouraged to prioritize participants with technical backgrounds for in-person registration.*** Individuals who are unable to register because the deadline has occurred or capacity has been reached will be added to a waitlist. Slots that remain open after registration close, or that become available due to cancellations will be filled on a first-come, first-served basis from the waitlist. An online registration form and various other meeting details can be found at the registration website, https://events.sa-meetings.com/VITALProposersDay

All registrants for the meeting who are not U.S. Citizens must complete and submit one of the following:

All U.S. Permanent Residents and Foreign Nationals must submit a DARPA Form 60, except foreign government personnel, who must submit only an Official Visit Request completed by their respective Embassy based in Washington, DC.

All forms must be submitted no later than **5:00 PM EDT on March 23, 2026**. Form 60 submission instructions are provided on the registration website and in the registration confirmation email. Contact your Embassy staff for assistance in submitting the Official Visit Request.

Contact Information:
Further administrative questions should be addressed to VITAL@darpa.mil . Please refer to the VITAL Proposers Day (DARPA-SN-26-55) in all correspondence. This announcement is not a request for abstracts or proposals; any sent will be disregarded.

DARPA hosts Proposers Days to provide potential performers with information on whether and how they might respond to the Government's research and development solicitations; and to increase efficiency in proposal preparation and evaluation. Therefore, Proposers Days are open only to potential proposers who have registered.

For those new to DARPA or national security, DARPA makes available a free, comprehensive resource via DARPAConnect on how to do business with the agency. In addition to DARPA 101 materials, relevant preparatory modules include "Making the Most of a Proposers Day" and "Understanding DARPA Broad Agency Announcements." Registration and access are free at www.darpaconnect.us.

Disclaimers:
This notice is issued solely for information and potential new program planning purposes; the SN does not constitute a formal solicitation for proposals or proposal abstracts. In accordance with FAR 15.201(e), responses to this notice are not offers and cannot be accepted by the Government to form a binding contract. Submission is voluntary and is not required to propose to subsequent Broad Agency Announcements (if any) or research solicitations (if any) on this topic. DARPA will not provide reimbursement for costs incurred in responding to this SN. Respondents are advised that DARPA is under no obligation to acknowledge receipt of the information received or provide feedback to respondents with respect to any information submitted under this SN.

NO CLASSIFIED INFORMATION SHOULD BE INCLUDED IN THE SN RESPONSE.